Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/08/2005
PI Name:	Takahashi, Fumiaki D.Eng.		
Project Title:	Physical and Chemical Aspects of Fire Suppression in Extraterestrial Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:	NOTE: Also affiliated with NASA Glenn Research Center		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/18/2000	**End Date:**	02/17/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Linteris, Gregory (National Institute of Standards and Technology) Katta, Viswanath (Innovative Scientific Solutions Inc.)		
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	In the exploration of space, longer duration missions to the moon, to Mars, and aboard the International Space Station (ISS) increase the likelihood of fire events. NASA's fire safety program for manned space flight is based largely upon controlling the flammability of the material on board and removing ignition sources. There is ongoing research to improve the flammability characterization of materials in low gravity. However, very little research has been conducted on fire suppression in a low-gravity environment. Although the existing suppression systems aboard the Space Shuttle (halon 1301, CF3Br) and the ISS (CO2 and water-based form) may continue to be used, alternative effective agents or techniques are desirable for long-duration missions. In this study, both experimental and computational approaches are pursued. The experiments include both normal- and reduced-gravity tests using the NASA Glenn's drop tower and the KC-135 aircraft. The critical extinguishment mole fraction of fire-extinguishing agents will be measured using a cup burner, which is the industry-standard burner		

	apparatus most widely used for agent effectiveness in fire safety engineering. The cup-burner forms a laminar co-flow diffusion flame with a circular fuel source (2.8 cm diameter, either a liquid pool or a low-velocity gas jet) inside a co-axial chimney with an oxidizer flow. Because its structure resembles a fire, great faith has been placed in agent extinction concentrations determined in the cup burner experiment, and many codes and standards are based on the cup-burner values. Major parameters to be varied are the gravity level, oxygen mole fraction in the oxidizer, oxidizer velocity, and types of fuel, agent, and diluent. Fuels include methane (CH4), heptane (n-C7H16), methanol, and trioxane (3[CH2O]). In addition to chemically passive agents (N2, CO2, Ar, and He) we will also study agents which have an increasing chemical contribution; (i.e., CF3H, CF3Br, and Fe(CO)5). Direct numerical simulation will be performed to capture unsteady fire suppression processes in various gravity and oxygen levels using comprehensive reaction mechanisms for the CH4-O2 combustion (GRI Mech) and halogenated agent chemistry (NIST CKMech).
Task Description:	The objectives of this study are as follows. 1. Obtain fundamental knowledge of physical and chemical processes of fire suppression, using gravity and oxygen concentration as independent variables to simulate various environments in spacecraft and surface bases in Mars and moon missions. 2. Provide rigorous testing of numerical models, which include comprehensive descriptions of chemistry and radiative heat losses. 3. Provide basic research results useful for technological advances in fire safety, including the development of new fire-extinguishing agents and approaches, in the microgravity environment associated with ISS and in the partial-gravity Martian and lunar environments. This research is important in relation to NASA's long-term goals in fire safety for long-duration missions in the exploration of space. First, this research contributes to the area of space fire suppression, which has not yet been fully studied. This research addresses such practical problems through fundamental scientific approaches. By using gravity and oxygen levels as independent variables, this research provides better understanding of the physical and chemical mechanisms involved in suppression of fires in space and gives more rigorous testing of numerical models. Basic results obtained by this research may lead to technological advances in fire suppression, e.g., new fire-extinguishing agents and/or approaches, particularly suitable for fires in space. Thus, this research contributes to the development of improved strategies and procedures for fire suppression in microgravity environment associated with the ISS or spacecraft and the reduced-gravity lunar and Martian environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the cup-burner apparatus has been used for determining the minimum extinguishing concentration of fire-extinguishing agents as described in national and international standards, fundamental understanding of this apparatus has been vertually null. This research provide the details of the flame extingushment process in the cup burner experiment, thus benefitting the terrestrial fire suppression engineering community and industry.
Task Progress:	The structure and suppression of laminar methane-air co-flow diffusion flames formed on a cup burner have been studied experimentally and numerically using physically acting fire-extinguishing agents (CO2, N2, He, and Ar) in normal earth (1g) and microgravity (ug). The low-g extinguishment testing of the flame with CO2 was conducted aboard the NASA Reduced-Gravity Aircraft (KC-135). The computation uses a direct numerical simulation with detailed chemistry and radiative heat-loss models. An agent was introduced into a low-speed coflowing oxidizing stream by gradually replacing the air until extinguishment occurred under a fixed minimal fuel velocity. The suppression of cup-burner flames, which resemble real fires, occurred via a blowoff process (in which the flame base drifted downstream) rather than the global extinction phenomenon typical of counterflow diffusion flames. The computation revealed that the peak reactivity spot (the reaction kernel) formed in the flame base was responsible for attachment and blowoff of the trailing diffusion flame. The minimum extinguishing concentration (MEC) determined experimentally in 1g were CO2, 15.7+/-0.6 %; N2, 25.9+/-1.0 %; He, 26.7+/-1.1 %; and Ar, 37.3+/-1.5 %. The extinguishing agent volume fraction in 1g for CO2 was nearly constant over a wide range of the oxidizer velocity (3 to 18 cm/s), exhibiting a “plateau region,” and steeply decreased as the velocity was decreased to approach the over-to-under-ventilated flame limit (~1 cm/s). The ug extinguishment data for CO2 acquired aboard the KC-135 were within 20 % to 24 %, but required further refinement. The numerical simulation performed thus far predicted the extinguishing agent volume fractions as: CO2, 14.5 % (or 16.1 % with different kinetic parameters for a methyl-H atom reaction step) in 1g and CO2, 19.1 %; N2, ~38 %; He, 30.7 %; and Ar, ~49 % in 0g. The buoyancy-induced flame flickering in 1g and thermal and transport properties of the agents affected the flame extinguishment limits.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Takahashi F, Linteris GT, Katta VR. "Fire Suppression in Low Gravity Using a Cup Burner." Strategic Research to Enable NASA's Exploration Missions Conference, Cleveland, Ohio, June 22-23, 2004. Abstracts of the Strategic Research to Enable NASA's Exploration Missions Conference, Cleveland, Ohio, June 22-23, 2004. NASA/TM-2004-213114. p. 165. NASA/CP—2004-213205/VOL2 , Jun-2004
Articles in Peer-reviewed Journals	Katta VR, Takahashi F, Linteris GT. "Suppression of cup-burner flames using carbon dioxide in microgravity." Combustion and Flame. 2004 Jun;137(4):506-22. https://doi.org/10.1016/j.combustflame.2004.01.015 , Jun-2004
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Presentation	Takahashi, F.; Linteris, G.; Katta, V. R. "Suppression Characteristics of Cup-Burner Flames in Low Gravity" AIAA 42nd Aerospace Sciences Meeting, Paper No. 2004-0957, Reno, Nevada Jan-2004
Presentation	Takahashi, F. "Space Experiment Concepts: Cup-Burner Flame Extinguishment" Conference-Workshop on Strategic Research to Enable NASA's Exploration Missions Cleveland, OH Jun-2004
Presentation	Takahashi, F.; Linteris, G.; Katta, V. R. "Cup-Burner Flame Extinguishment in Low Gravity" Thirtieth International Symposium on Combustion, Chicago, IL Jul-2004